

Shane Paton

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Clearance: Interim Secret (TS/SCI in progress)

Education

Miami University, Oxford, OH
Major: Computer Science | Minor: Management
GPA: 3.98 / 4

Expected 12/2026

Skills

Languages: C++, Python, Java, Dart, JavaScript, TypeScript, SQL

Technologies: React, Next.js, Flutter, Firebase, PostgreSQL, Azure, AKS, Kubernetes, Git, GitHub Actions, Docker, Jira

Work Experience

84.51° - Software Engineering Intern 05/2026 – 08/2026

- Developed an LLM observability platform supporting client-specific analysis of 10,000+ prompts using embeddings, HDBSCAN, and LLMs for intent clustering and theme generation
- Deployed the application on Azure using AKS, PostgreSQL, Key Vault, Helm, and Terraform, implementing authentication and Microsoft Graph API email integration
- Automated RAID log management by extracting items from Teams meetings and updating the log in Power Automate

Modern Technology Solutions, Inc. (MTSI) - Software Engineering Intern 05/2025 – 08/2025

- Synchronized simulations across 5+ networked devices by implementing Distributed Interactive Simulation (DIS) protocols over UDP
- Simulated realistic object behavior with a custom physics engine supporting 100+ moving objects at 60 FPS
- Reduced file size by up to 60% with a custom Huffman compressor/decompressor in C++, visualized decompression and the Huffman Tree using MIXR

Noshable - Website Developer Intern 05/2024 – 08/2024

- Enhanced data visualization by building a dynamic map page using Google Maps API, displaying over 9,000 custom markers to help efficiently organize cleaner availability and client onboarding
- Utilized Firebase for cloud storage to streamline datasets and reduce unnecessary API calls, saving ~\$2,000 monthly

Projects

Swing Coach - Next.js, TypeScript, MediaPipe, Gemini

- Built a real-time golf coaching app using MediaPipe pose to extract setup, kinematic, swing path, and stability metrics
- Improved capture reliability with auto-recording, framing checks, stillness validation, and swing-end detection
- Generated Gemini coaching feedback using reference ranges, metric weighting, and prompt guardrails

Sports Modeling & Analytics Club Platform - Next.js, TypeScript, Supabase, PostgreSQL

- Built a full-stack sports analytics platform for ~50 members to publish articles and manage betting insights
- Implemented authentication, role-based access control, and real-time data synchronization using Supabase
- Developed interactive features for article voting, threaded comments, user picks, and SMAC coin tracking

Spotify Song Ranker - React

- Introduced song searching via Spotify API, enabling users to fetch tracks, streamlining user exploration & interaction
- Incorporated real-time UI updates by managing state with React hooks, improving responsiveness & user interaction
- Implemented drag-and-drop song ranking with React, replicating Spotify's design for a familiar and intuitive UX

Leadership & Involvement

Sports Modeling & Analytics Club - Co-founder & CTO 05/2025 – Present

- Co-founded a student organization focused on sports analytics, predictive modeling, and responsible betting research

Techstars Startup Weekend 08/2023

- Collaborated with a 5-person team to rapidly prototype and pitch a startup, delivering a VC-style presentation to ~100 attendees; generated \$1,000 in revenue and placed 3rd, winning an additional \$1,000

Memberships & Honors

Association for Computing Machinery (ACM) - Club | 02/2025 – Present

Tau Beta Pi - Engineering Honor Society | 2025 – Present

Theta Tau - Professional Engineering Fraternity | 2025 – Present